A GOOD RESOLUTION COMPUTES THE COHOMOLOGY OF THE TOTAL COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- The short exact sequences relating kernels, images and cohomology of a complex. Standard: for a complex of abelian groups the kernel, the image and the cohomology fit into the two displayed short exact sequences. Supplies the exact sequences used to propagate exactness from kernels to images and then to cohomology.

Lemma 0.1. *Let $M^\bullet$ be a complex of abelian groups. Let*

$$\ldots \to A_2^\bullet \to A_1^\bullet \to A_0^\bullet \to M^\bullet \to 0$$

be an exact complex of complexes of abelian groups such that for all $p \in \mathbf{Z}$ the complexes

$$\ldots \to \mathrm{Ker}(d^p_{A_2^\bullet}) \to \mathrm{Ker}(d^p_{A_1^\bullet}) \to \mathrm{Ker}(d^p_{A_0^\bullet}) \to \mathrm{Ker}(d^p_{M^\bullet}) \to 0$$

are exact as well. Set $A^{p,q} = A^q_{-p}$ to obtain a double complex. Then $Tot(A^{\bullet,\bullet}) \to M^\bullet$ induced by $A_0^\bullet \to M^\bullet$ is a quasi-isomorphism.

Proof. Using the short exact sequences $0 \to \mathrm{Ker}(d^p_{A_n^\bullet}) \to A^p_n \to \mathrm{Im}(d^p_{A_n^\bullet}) \to 0$ and the assumptions we see that

$$\ldots \to \mathrm{Im}(d^p_{A_2^\bullet}) \to \mathrm{Im}(d^p_{A_1^\bullet}) \to \mathrm{Im}(d^p_{A_0^\bullet}) \to \mathrm{Im}(d^p_{M^\bullet}) \to 0$$

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is exact for all $p \in \mathbf{Z}$. Repeating with the exact sequences $0 \to \mathrm{Im}(d^{p-1}_{A^\bullet_n}) \to \mathrm{Ker}(d^p_{A^\bullet_n}) \to H^p(A^\bullet_n) \to 0$ we find that

$$\ldots \to H^p(A^\bullet_2) \to H^p(A^\bullet_1) \to H^p(A^\bullet_0) \to H^p(M^\bullet) \to 0$$

is exact for all $p \in \mathbf{Z}$.

Write $T^\bullet = \mathrm{Tot}(A^{\bullet,\bullet})$. We will show that $H^0(T^\bullet) \to H^0(M^\bullet)$ is an isomorphism. The same argument works for other degrees. Let $x \in \mathrm{Ker}(\mathrm{d}^0_{T^\bullet})$ represent an element $\xi \in H^0(T^\bullet)$. Write $x = \sum_{i=n,\ldots,0} x_i$ with $x_i \in A^i_i$. Assume $n > 0$. Then x_n is in the kernel of $d^n_{A^\bullet_n}$ and maps to zero in $H^n(A^\bullet_{n-1})$ because it maps to an element which is the boundary of x_{n-1} up to sign. By the first paragraph of the proof, we find that $x_n \bmod \mathrm{Im}(d^{n-1}_{A^\bullet_n})$ is in the image of $H^n(A^\bullet_{n+1}) \to H^n(A^\bullet_n)$. Thus we can modify x by a boundary and reach the situation where x_n is a boundary. Modifying x once more we see that we may assume $x_n = 0$. By induction we see that every cohomology class ξ is represented by a cocycle $x = x_0$. Finally, the condition on exactness of kernels tells us two such cocycles x_0 and x_0' are cohomologous if and only if their image in $H^0(M^\bullet)$ are the same. $\square$